4CBLB20: High-Precision Robot Arm Positioning Using Feedback and Feedforward Control

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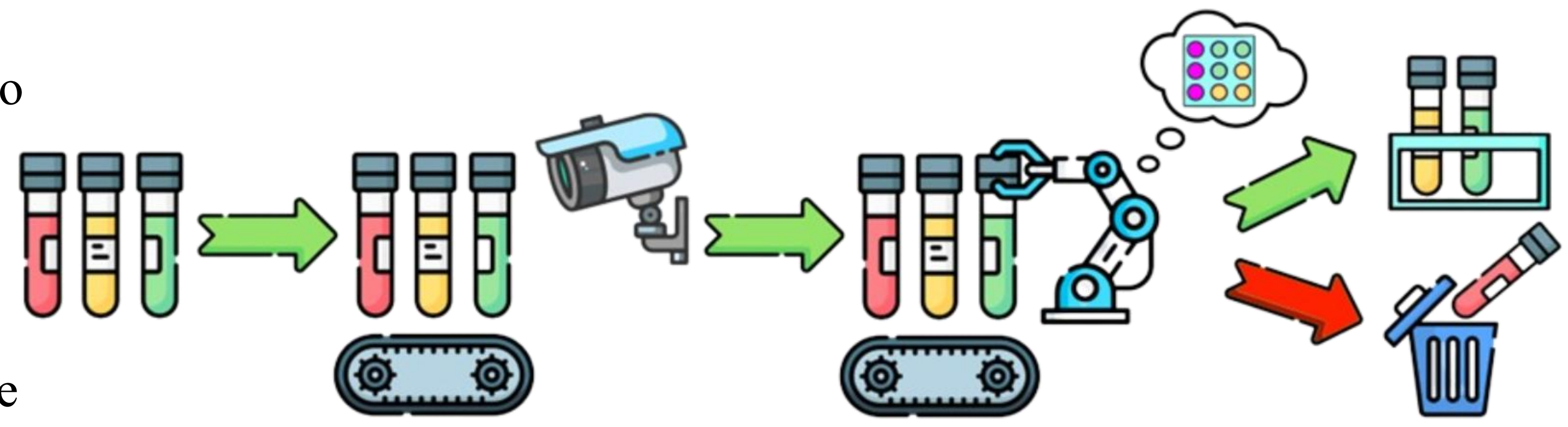
The Task:

- To cut out human contact with hazardous vials through a robotic arm.
- The robot is fed a .png image of a 3x3 color grid, which is translated into a positioning matrix.
- The robot arm is fed colored vials (indicating positive/negative results) via a conveyor.
- The vial is picked up and based on the color the vial is assigned a storage position within the grid or disposed of. Useful for lab testing like filling a PCR machine.



Controller design: Feedback & Feedforward

Figure 2 shows the control scheme of the system. To design the controller the FRF is analysed and countermeasures are taken. The analyses for the R-axis follows.

FRF Issue	Details	Counter Measure
Very Low Magnitude	(−10 dB to −100 dB): Arm lacked stiffness or had high inertia.	**PI+PD controller** boosted low-frequency response with a D-value of **0.5** and a gain of **1**.
Resonance	(80–110 Hz): Sharp magnitude peak risked instability.	**Notch filter** (**80–130 Hz**) tamed the resonance. With $\boldsymbol{\beta_1 = 0.5}$ **&** $\boldsymbol{\beta_2 = 0.2}$.
High-Frequency noise	Noise could be amplified by derivative action.	**Low-pass filter (100 Hz)** reduced high-frequency gain.
Stiction	(1–6 Hz): Static friction caused sluggish low-frequency behavior.	**PI zero** at **5 Hz** improved steady-state accuracy.

Giving a bandwidth of **19.1 Hz**, MM of **4.64 dB**, PM of **48.2º** and GM of **16 dB**.

The feedforward gains associated with the coulomb, viscous and acceleration terms were determined by means of trial and error. Optimizing for the error found in Figure 3 for a full motion sweep back and forth. As a starting point the coulomb value has been determined with a ramp function until movement occurred. Leading to a **Kc of 0.094**, **Kv of 0.005** and **Kacc of 0.0005.**

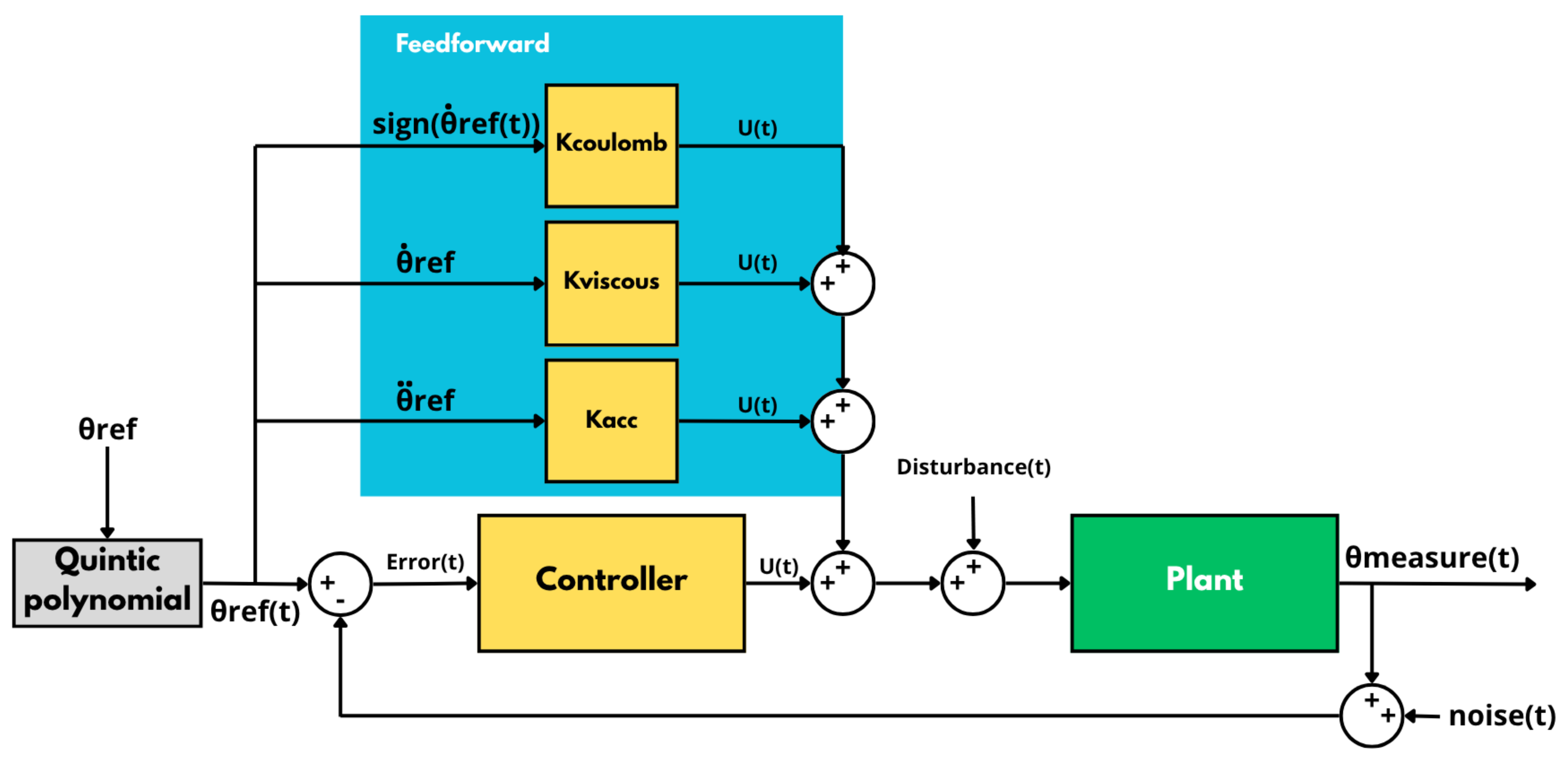


Figure 2: Control block diagram

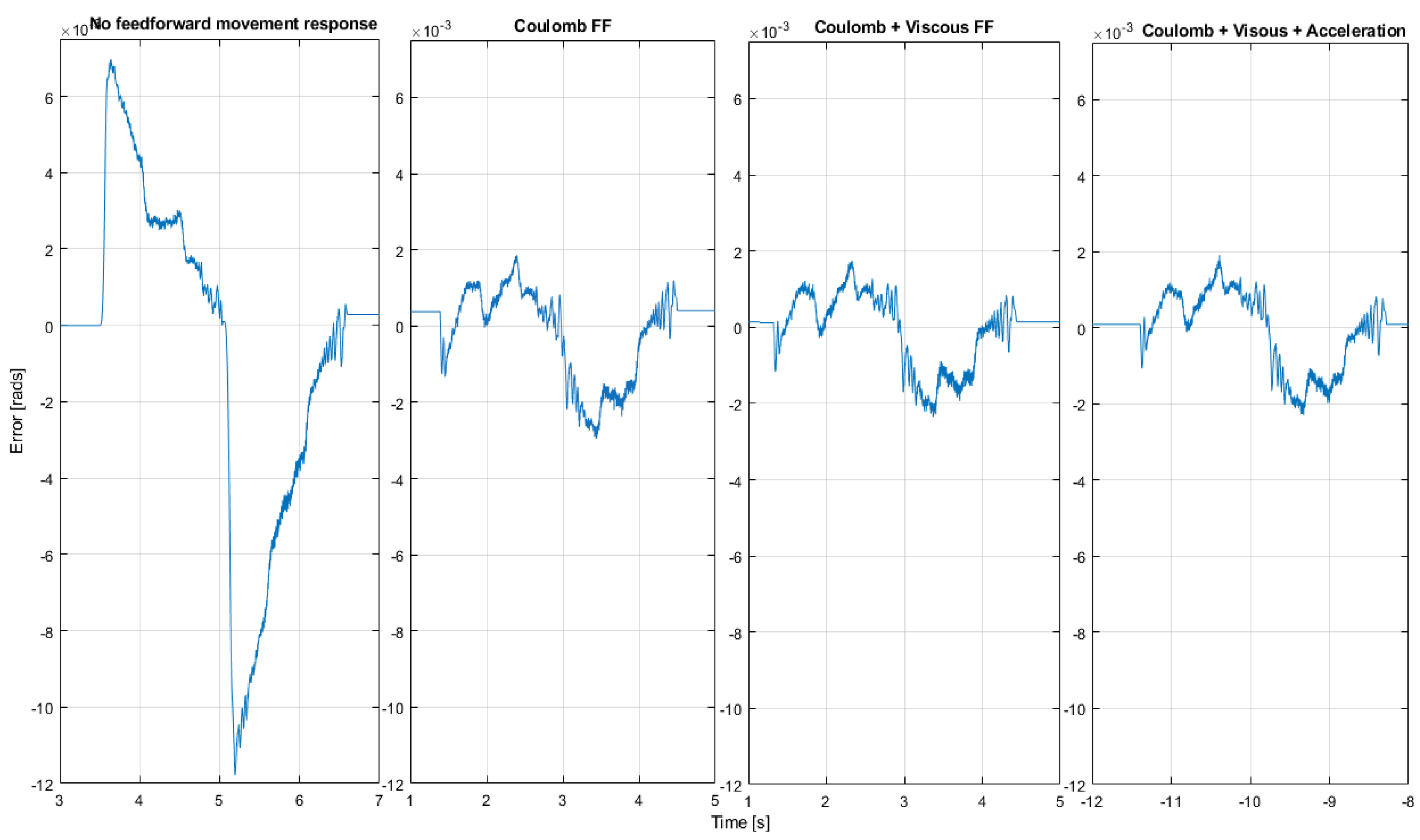


Figure 3: The error function of the R coordinate for a full sweep using the final controller and FF parts.

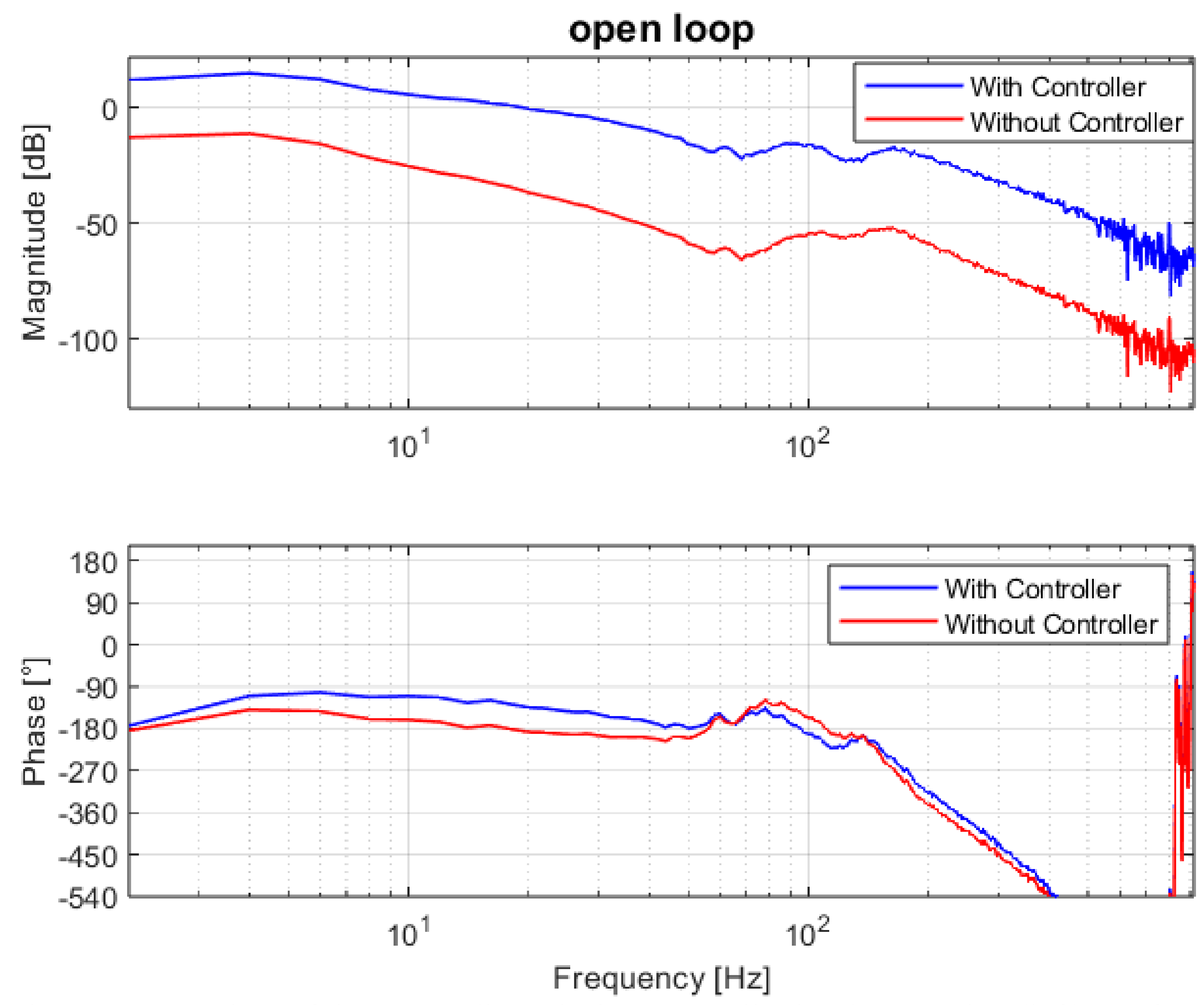


Figure 1. FRF Plant of the R coordinate before controller implementation (left) vs after(right) .

System Requirement Priorities:

1. Accuracy and precision

To avoid breakage of vials and wrong placement of test results, pick-and-place accuracy and smooth, gentle movements were the top priority. An accuracy of **2 mm** is aimed for with a pick-up centering accuracy of **1 mm**.

2. Speed

Quickly sorting vials to speed up test results was the second priority of the system. Aiming for a Pick and place cycle within **10 seconds**.

3. Robustness

Controller Robustness was the priority coming last in our design of the robot. This will be achieved by having a bandwidth of at least **15 Hz**, a MM < **6dB**, a PM > **30º** and a GM > **6 dB**.

Performance evaluation

Pick-up accuracy [mm]
X-axis: 2.6 ± 1.1
Y-axis: 3.2 ± 1.3

Mean placement error [mm]
X-axis: 0.0175
Y-axis: 0.135
Z-axis: 0.286

Pick-place time
13 seconds

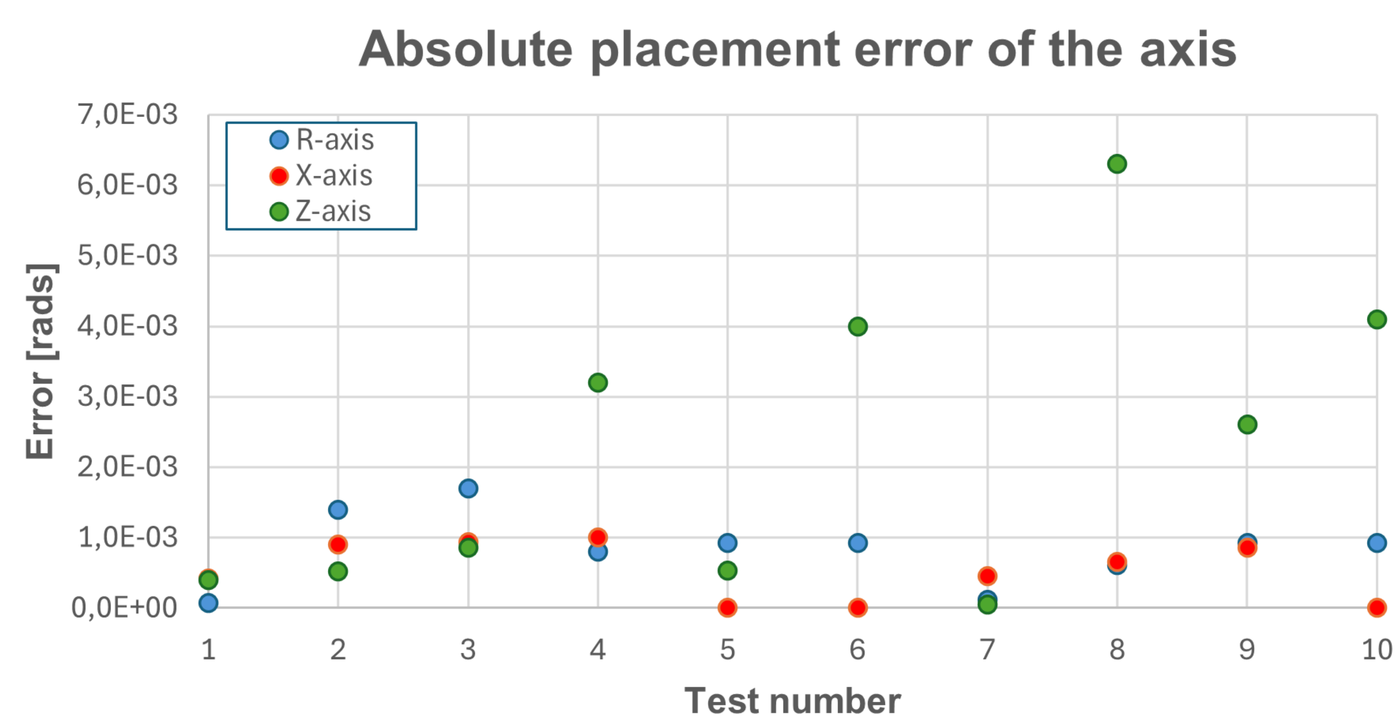


Figure 4: Final placement error of the separate axis.